CPU SCHEDULING

1. What is an operating system?

An operating system is a program that manages the computer hardware. It also provides a basis for application program and acts as an intermediary between a user of a computer and the computer hardware.

2. What is meant by timesharing?

Timesharing (or multitasking) is a logical extension of multiprogramming. The CPU executes multiple jobs by switching among them, but the switches occurs so frequently that the users can interact with each program while it is running.

3. What are the advantages of a time-sharing operating system?

The advantages of a time-sharing operating system are:a.It allows many users to share the computer simultaneously.b.The use of virtual memory frees programmers from concern over memory-storage limitations.

4. When are real-time systems used?

A real-time system is used when rigid time requirements have been placed on the operation of a processor or the flow of data. It is often used as a control device in a dedicated application.

5. What are system calls?

System calls provide the interface between a process and the operating system. These calls are generally available as assembly-language instructions, and they are usually listed in various manuals used by the assembly-language programmers.Certain systems allow system calls to be made directly from a higher level language program, in which case the calls normally resemble predefines function or subroutine calls.

6. What are daemon processes?

Processes that stay in the background to handle some activities are called daemon processes. Eg., processes in background to handle activity such as email, web pages, news, printing, etc.

7. What is meant by CPU scheduler?

Whenever the CPU becomes idle, the operating system must select one of the processes in the ready queue to be executed. The selection process is carried out by the short-term-scheduler (or CPU scheduler).

8. What is medium term scheduler?

It removes processes from memory (and from active contention for the CPU), and thus reduces the degree of multiprogramming. At some later time, the process can be reintroduced into memory and its execution can be continued where it left out.(swapping)

9. What is long term scheduler?

Long term scheduler or job scheduler, selects from among the processes from this pool loads them into memory for execution.

10. What is meant by degree of multiprogramming?

The long-term scheduler controls the degree of multiprogramming – the number of processes in memory.

11. What are the two types of scheduling? Explain.

The two types of scheduling are:a.Preemptive schedulingb.Nonpreemptive scheduling Preemptive scheduling is the process of scheduling in which, a process can run for a maximum of fixed time. Once the end of time interval is reached, it is suspended and the next process runs.Nonpreemptive scheduling is the process of scheduling in which, a process runs until it completes its operation or it voluntarily releases the CPU.

12. What is a dispatcher?

The dispatcher is the module that gives control of the CPU to the process selected by the short-term-scheduler. This function involves:a.Switching contextb.Switching to user modec.Jumping to proper location in the user program to restart that program.

13. What is meant by dispatch latency?

The time taken for the dispatcher to stop one process and start another running is known as dispatch latency.

14. List the various scheduling criterions.

The scheduling criterions include :

a.CPUutilization b.Throughput c.Turnaround time d.Waiting time e.Response time

15. Define throughput.

The number of processes completed per time unit is called throughput.

16. Define turnaround time.

The interval from the time of submission of a process to the time of completion is the turnaround time.

17. Define waiting time.

Waiting time is the sum of the periods spent waiting in the ready queue.

18. Define response time.

Response time is the amount of time it takes to starts responding.

19. Explain multilevel queue scheduling.

A multilevel queue scheduling algorithm partitions the ready queue into several separate queues. The processes are permanently assigned to one queue, generally based on some property of process, such as memory size, process priority, or process type. Each queue has its own scheduling algorithm. For example, separate queue might be used for foreground and background processes. The foreground queue might be scheduled by an RR algorithm, while the background queue is scheduled by an FCFS algorithm.

20. How are processes scheduled in multilevel feedback queue scheduling?

Multilevel feedback queue scheduling, allows a process to move between queues. The idea is to separate processes with different CPU-burst characteristics. If a process uses too much CPU time, it will be moved to a lower-priority queue. This scheme leaves I/O-bound and interactive processes in the higher-priority queues. Similarly, a process that waits too long in a lower-priority queue may be moved to a higher-priority queue. This form of aging prevents starvation.

21. What is context switch?

Switching the CPU to another process requires saving the state of the old process and loading the saved state for the new process. This task is known as a context switch.

22. What is thread?

A thread sometimes called as light weight process (LWP), is a basic unit of CPU utilization; it comprises a thread id, a program counter, a register set, and a stack.

23. What is heavyweigh thread?

A traditional (or heavyweight) process has a single thread of control

PROCESS SYNCHRONIZATION

1. What is a process?

A process is a program in execution. A process is more than the program code, which is sometimes known as the text section. It also includes the current activity, as represented by the value of the program counter and the contents of the processor's registers.

2. What is a child process?

A process may create several new processes, via a create-process system call, during the course of execution. The creating process is called a parent process, where as the new processes are called the children of that process.

3. What are the various states of process?

As a process executes, it changes state. The state of a process is defined in part by the current activity of that process. Each process may be in one of the following states:a.New: The process being created.b.Running: Instructions are being executed.c.Waiting: The process is waiting for some event to occur (such as an I/O completion or reception of a signal).d.Ready: The process is waiting to be assigned to a processor.e.Terminated: The process has finished execution.

4. What is a process control block?

Each process is represented in the operating system by a process control block (PCB) – also called a task control block. It contains many pieces of information associated with a specific process, including these:a.Process stateb.Program counterc.CPU registersd.CPU-scheduling informatione.Memory-management informationf.Accounting informationg.I/O status information

5. Explain CPU-bound and I/O-bound processes.

CPU-bound processes are those that spend most of their time in computer. They have long CPU bursts.I/O-bound processes are those that spend most of their time waiting for I/O. They have short CPU bursts.

6. What is a ready queue?

The processes that are residing in main memory and are ready and waiting to execute are kept on a list called the ready queue. This queue is generally stored as a linked list.

7. What is meant by inter process communication?

IPC provides a mechanism to allow processes to communicate and to synchronize their actions without sharing the same address space. IPC is particularly useful in a distributed environment where the communicating processes may reside on different computers connected with a network.

8. What is prefetching?

It is a method of overlapping the I/O of a job with that job's own computation. After read operation completes and the job is about to start operating on the data, the input device is instructed to begin the next read immediately.

9. What is process synchronization?

Process synchronization refers to the idea that multiple processes are to join up or handshake at a certain point, in order to reach an agreement or commit to a certain sequence of action.

10. What is critical section?

Consider a system consisting of *n* processes *{Po,P1, ..., P,-1)*. Each process has a segment of code, called a critical section, in which the process may be changing common variables, updating a table, writing a file, and so on. The important feature of the system is that, when one process is executing in its critical section, no other process is to be allowed to execute in its critical section. Thus, the execution of critical sections by the processes is *mutually exclusive* in time.

11. Structute of critical section?

do

{

Entry section

Critical section

Exit section

Remainder section

} while(1)

12. What are the three requirements to satisfy critical section?

Mutual Exclusion:

Progress

Bounded Waiting

13. What is mutual exclusion

If process *Pi* is executing in its critical section, then no other processes can be executing in their critical sections.

14. What is progress?

If no process is executing in its critical section and some processes wish to enter their critical sections, then only those processes that are not executing in their remainder section can participate in the decision on which will enter its critical section next, and this selection cannot be postponed indefinitely.

15. What is bounded waiting?

There exists a bound on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted.

16. Structure of Two process solution for algorithm 1?

```
do
{
while(turn !=1)
Critical section
turn=j;
Remainder section
} while(1)
```

```
if (S.value < 0) {
add this process to S . L;
block( );
}
}
```

22. Modification definition of signal semaphore operations.

```
void signal(semaphore S) {
S.value++;
if (S.value <= 0) {
remove a process P from S . L ;
wakeup (P) ;
}
}
```

23. What is semaphore?

Semaphore is a synchronization tool that can resolve complex situation of critical section problems. A semaphore S is an integer variable that, apart from initialization, is accessed only through two standard atomic operations: wait and signal.

24. pseudocode for wait operation

```
wait(S) {
while (S <= 0)
; // no-op
S--;
}
```

25. pseudocode for signal operation

```
Signal(S){
S++;
```

DEADLOCK

1. What is deadlock?

A process requests resources; if the resources are not available at that time, the process enters a wait state. Waiting processes may never again change state, because the resources they have requested are held by other waiting processes. This situation is called a deadlock.

2. Conditions to avoid deadlock?

Mutual exclusion: At least one resource must be held in a non sharable mode; that is, only one process at a time can use the resource. If another process requests that resource, the requesting process must be delayed until the resource has been released.

Hold and wait: A process must be holding at least one resource and waiting to acquire additional resources that are currently being held by other processes.

No preemption: Resources cannot be preempted; that is, a resource can be released only voluntarily by the process holding it, after that process has completed its task.

Circular wait: A set {Po, PI, ..., Pn,) of waiting processes must exist such that Po is waiting for a resource that is held by PI, PI is waiting for a resource that is held by P2, ..., P3, PI is waiting for a resource that is held by Pn,, and Pn, is waiting for a resource that is held by Po.

3. What is resource allocation graph?

Deadlocks can be described more precisely in terms of a directed graph called a system resource-allocation graph. This graph consists of a set of vertices V and a set of edges E. The set of vertices V is partitioned into two different types of nodes P = {PI, P2, ..., Pn}, the set consisting of all the active processes in the system, and R = {R1, R2, ..., Rm), the set consisting of all resource types in the system.

A directed edge from process Pi to resource type Rj is denoted by Pi -> Rj; it signifies that process Pi requested an instance of resource type Ri and is currently waiting for that resource.A directed edge from resource type Ri to process Pi is denoted by Rj -> Pi; it signifies that an instance of resource type Ri has been allocated to process Pi. A directed edge ->is called a **request edge**; a directed edge ->is called an **assignment edge**.

4. What is wait-for graph?

An edge from Pi to ***Pi*** in a wait-for graph implies that process Pi is waiting for process Pi to release a resource that Pi needs. **An** edge ***Pi*** ->*Pj* exists in a wait-for graph if and only if the corresponding resource allocation graph contains two edges ***Pi*** -> *Rq* and *Rq* ->*Pj* for some resource *Rq*. A deadlock exists in the system if and only if the wait-for graph contains a cycle. To detect deadlocks, the system needs to *maintain* the wait-for graph and periodically to *invoke an algorithm* that searches for a cycle in the graph.

5. How can we handle deadlock?

Deadlock can be handled in one of the following ways:a.Protocols can be used to prevent or avoid deadlocks, ensuring that the system will never enter a deadlock state.b.The system can be allowed to enter a deadlock state which can be detected and recovered.c.The problem can be ignored and pretending that deadlocks never occurs in the system.

6. What is meant by deadlock prevention?

Deadlock prevention is a set of methods for ensuring that at least one of the necessary conditions cannot hold.

7. Write notes on deadlock avoidance.

Deadlock avoidance; require that the operating system be given in advance additional information concerning which resources a process will request and use during its lifetime.

8. Recovery from deadlock

Deadlock prevention methods

a. Abort all deadlocked processes

b. Abort one process at a time until the deadlock cycle is eliminated:

Resource preemption

a. Selecting a victim

b. Starvation

c. Rollback

9. Define Rollback: If we preempt a resource from a process, what should be done with that process? Clearly, it cannot continue with its normal execution; it is missing some needed resource. We must roll back the process to some safe state, and restart it from that state.

10. Define Selecting a victim: Which resources and which processes are to be preempted? As in process termination, we must determine the order of pre-emption to minimize cost. Cost factors may include such parameters as the number of resources a deadlock process is holding, and the amount of time a deadlocked process has thus far consumed during its execution.

11. Define Starvation: How do we ensure that starvation will not occur? That is, how can we guarantee that resources will not always be preempted from the same process? As a result, this process never completes its designated task, a starvation situation that needs to be dealt with in any practical system. Clearly, we must ensure that a process can be picked as a victim only a (small) finite number of times.

1. Define Address Binding

A program resides on a disk as a binary executable file. Depending on the memory management in use, the process may be moved between disk and memory during its execution. The collection of processes on the disk that is waiting to be brought into memory for execution forms the input queue.

2. Define Compile time: If you know at compile time where the process will reside in memory, then absolute code can be generated

3. Define Load time: If it is not known at compile time where the process will reside in memory, then the compiler must generate relocatable code. In this case, final binding is delayed until load time. If the starting address changes, we need only to reload the user code to incorporate this changed value.

4. Define Execution time: If the process can be moved during its execution from one memory segment to another, then binding must be delayed until run time.

5. Define logical address & physical address?

An address generated by the CPU is commonly referred to as a logical address, whereas an address seen by the memory unit-that is, the one loaded into the memory-address register of the memory-is commonly referred to as a physical address.

6. Define virtual address

The compile-time and load-time address-binding methods generate identical logical and physical addresses. However, the execution-time address binding scheme results in differing logical and physical addresses. In this case, we usually refer to the logical address as a virtual address.

7. Define dynamic linking?

Rather than loading being postponed until execution time, linking is postponed. This feature is usually used with system libraries, such as language subroutine libraries.

Without this facility, all programs on a system need to have a copy of their language library (or at least the routines referenced by the program) included in the executable image.

8. What is overlays

To enable a process to be larger than the amount of memory allocated to it, we can use overlays. The idea of overlays is to keep in memory only those instructions and data that are needed at any given time. When other instructions are needed, they are loaded into space occupied previously by instructions that are no longer needed.

9. What is contiguous memory allocation?

In this contiguous memory allocation, each process is contained in a single contiguous section of memory.

10. Define holes and blocks?

Initially all memory is available for user process and is considered one large block of available memory, a hole.

11. Explain the memory allocation algorithms?

First fit: Allocate the first hole that is big enough. Searching can start either at the beginning of the set of holes or where the previous first-fit search ended. We can stop searching as soon as we find a free hole that is large enough.

Best fit: Allocate the smallest hole that is big enough. We must search the entire list, unless the list is kept ordered by size. This strategy produces the smallest leftover hole.

Worst fit: Allocate the largest hole. Again, we must search the entire list, unless it is sorted by size. This strategy produces the largest leftover hole, which may be more useful than the smaller leftover hole from a best-fit approach.

12. Brief about 50-percent rule?

Statistical analysis of first fit, for instance, reveals that, even with some optimization, given N allocated blocks, another *0.5N* blocks will be lost due to fragmentation. That is, one-third of memory may be unusable! This property is known as the 50-percent rule.

13. Define internal fragmentation?

It is the space waste inside of allocated memory on the allowed sizes of allocated blocks

14. Define external fragmentation?

It is the phenomenon in which free storage become divided into many small pieces over time. It occurs when an application allocates & de-allocates regions of storage of varying sizes & the allocation algorithm responds by leaving the allocated & de-allocated regions interspersed. The result in that although free space is available it is effectively unusable because it is divided into pieces that are too small to satisfy the demands of the application.

15. One solution to the problem of external fragmentation is compaction. The goal is to shuffle the memory contents to place all free memory together in one large block.

16. Define Paging

Paging is a memory-management scheme that permits the physical-address space of a process to be noncontiguous. Physical memory is broken into fixed-sized blocks called frames. Logical memory is also broken into blocks of the same size called pages. Every address generated by the CPU is divided into two parts: a page number (p) and a page offset (d). The page number is used as an index into a page table. The page table contains the base address of each page in physical memory.

17. Define TLB?

The TLB is associative, high-speed memory. Each entry in the TLB consists of two parts: a key (or tag) and a value. When the associative memory is presented with an item, it is compared with all keys simultaneously. If the item is found, the corresponding value field is returned. The search is fast; the hardware, however, is expensive.

18. Define segmentation?

Segmentation is one of the most common ways to achieve memory protection. In a computer system using segmentation, an instruction operand that refers to a memory location includes a value that identifies a segment and an offset within that segment.

19. What is the difference between Paging and Segmentation?

In paging, memory is divided in to equal size segments called pages whereas memory segments could vary in size (this is why each segment is associated with a length attribute). Sizes of the segments are determined according to the address space required by a process, while address space of a process is divided in to pages of equal size in paging. Segmentation provides securityassociated with the segments, whereas paging does not provide such a mechanism.

20. Define demand segmentation ?

It can also be used to provide virtual memory. However, segment-replacement algorithms are more complex than are page-replacement algorithms because the segments have variable sizes.

21. Define demand Paging?

A demand-paging system is similar to a paging system with swapping . Processes reside on secondary memory (which is usually a disk). When we want to execute a process, we swap it into memory. Rather than swapping the entire process into memory, however, we use a lazy swapper. A lazy swapper never swaps a page into memory unless that page will be needed.

22. Explain about Zero-fill-on-demand pages

Zero-fill-on-demand pages have been zeroed-out before being allocated, thus erasing the previous contents of the page. With copy-on-write, the page being copied will be copied to a zero-filled page.

PAGE-REPLACEMENT ALGORITHMS

1. Define frame-allocation algorithm and a page-replacement algorithm

   If we have multiple processes in memory, we must decide how many frames to allocate to each process. Further, when page replacement is required, we must select the frames that are to be replaced.

2. What is meant by memory reference?

   The string of memory references to computing the number of page faults is called a reference string.

3. Define FIFO page replacement?

   A FIFO replacement algorithm associates with each page the time when that page was brought into memory. When a page must be replaced, the oldest page is chosen.

4. Explain Belady's anomaly?

   For some page-replacement algorithms, the pagefault rate may *increase* as the number of allocated frames increases.

5. Define Optimal page replacement?

   It is simply to replace the page that will not be used for the longest period of time.

6. Define LRU page replacement?

   The key distinction between the FIFO and OPT algorithms (other than looking backward or forward in time) is that the FIFO algorithm uses the time when a page was brought into memory; the OPT algorithm uses the time when a page is to be *used.* If we use the recent past as an approximation of the near future, then we will replace the page that *has not been used* for the longest period of time.

7. Discuss about Second-Chance Algorithm

   The basic algorithm of second-chance replacement is a FIFO replacement algorithm. When a page has been selected, however, we inspect its reference bit. If the value is 0, we proceed to replace this page. If the reference bit is set to 1, however, we give that page a second chance and move on to select the next FIFO page. When a page gets a second chance, its reference bit is cleared and its arrival time is reset to the current time. Thus, a page that is given a second chance will not be replaced until all other pages are replaced.

8. Brief Enhanced Second-Chance Algorithm

FILE ALLOCATION

1. List the file allocation attributes.

Name: The symbolic file name is the only information kept in human readable form.

Identifier: This unique tag, usually a number, identifies the file within the file system; it is the non-human-readable name for the file.

Type: This information is needed for those systems that support different types.

Location: This information is a pointer to a device and to the location of the file on that device.

Size: The current size of the file (in bytes, words, or blocks), and possibly the maximum allowed size are included in this attribute.

Protection: Access-control information determines who can do reading, writing, executing, and so on.

2. Define sequential access

The simplest access method is sequential access. Information in the file is processed in order, one record after the other.

A file is made up of fixed length logical records that allow programs to read and write records rapidly in no particular order.

3. Define boot control block.

A **boot control block** can contain information needed by the system to boot an operating from that partition. If the disk does not contain an operating system, this block can be empty. It is typically the first block of a partition.In UFS, this is called the **boot block;** in NTFS, it is the **partition boot sector.** A **partition control block** contains partition details, such as the number of blocks in the partition, size of the blocks, free-block count and free-block pointers, and free FCB count and FCB pointers. In UFS this is called a **superblock;** in NTFS, it is the **Master File Table.**

4. Define system wide open file table?

10. What is meant by Grouping?

Rather than keeping a list of n free disk addresses, we can keep the address of the first free block and the number n of free contiguous blocks that follow the first block. Each entry in the free-space list then consists of a disk address and a count.

11. What is spooling?

A **spool** is a buffer that holds output for a device, such as a printer, that cannot accept interleaved data streams. Although a printer can serve only one job at a time, several applications may wish to print their output concurrently, without having their output mixed together.

12. What's the difference between Buffering and Caching?

Even though both caching and buffering involves storing data temporally in a different location, they have some important differences. Caching is done to reduce the access time in retrieving data from a slower storage device. It is based on the principle that the same data will be accessed multiple times thus storing them in cache would reduce the access time largely. Buffering is mainly used to overcome the difference between the speeds in which the data is received and data is processed by a device.

DISK SCHEDULING

1. Define seek time

   The seek time is the time for the disk arm to move the heads to the cylinder containing the desired sector.

2. What is rotational latency?

   The rotational latency is the additional time waiting for the disk to rotate the desired sector to the disk head.

3. Define bandwidth

   The disk bandwidth is the total number of bytes transferred, divided by the total time between the first request for service and the completion of the last transfer. We can improve both the access time and the bandwidth by scheduling the servicing of disk I/O requests in a good order.

   The simplest form of disk scheduling is, of course, the first-come, first-served (FCFS) algorithm. This algorithm is intrinsically fair, but it generally does not provide the fastest service. The request which comes first will be served

4. Concept of SSTF ?

   The SSTF algorithm selects the request with the minimum seek time from the current head position. Since seek time increases with the number of cylinders traversed by the head, SSTF chooses the pending request closest to the current head position.

5. Breif about SCAN

   In the SCAN algorithm, the disk arm starts at one end of the disk, and moves toward the other end, servicing requests as it reaches each cylinder, until it gets to the other end of the disk. At the other end, the direction of head movement is reversed, and servicing continues. The head continuously scans back and forth across the disk. The SCAN algorithm is sometimes called the elevator algorithm, since the disk arm behaves just like an elevator in a building, first servicing all the requests going up, and then reversing to service requests the other way.

LINUX

1. What is linux?

   Linux is a Unix like operating system which was developed in 1991 by Tinnish student called LINUX TORVALDS.

2. Write about cut command in linux

   Cut command is used to select section of text from each line of files.

   Cat file.txt

   Unix or linux os

   Cut –c4 file.txt

   o/p: x

3. Define AWK

   Awk –Aho Weinberger and Kernighan, AWK is an interpreted programming language which focuses on processing text. It make heavy use of strings , arrays and regular expressions and is usefull for parsing system data and generating automatic reports.

4. Gwak is a powerful pattern matching and processing language.it is based on AWK.

   -f program file: read the AWKprogram source from the file program file, instead of from the first command line argument.

4. Define GREP.

   GREP: GLOBAL REGULAR EXPRESSION PRINT

   The grep commands allows you to search one file or multiple files for lines contain apattern.

   Exit status is 0 if matches were found , and 2 if error occur

5. Write the features of linux :

   Multiprogramming

   Time sharing

   Multi tasking

   Virtual memory

   Shared libraries

6. What are the components of a linux system: